

# Capital Goods & End-of-Life (Machinery) in LCA - Modelling Approach, Calculation Framework and Default Values

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**Domain of application:** Textile manufacturing — Knitting (circular, flat, seamless, hosiery, 3D/WHOLEGARMENT), and any capital-intensive industrial process with identifiable machinery.

**Version:** 1.0 - February 2025

**Supporting model:** [DatabaseLCI based on referencedata.xlsx](#)

## 1. Purpose and Scope

This document describes the methodological framework for including **machinery capital goods and their end-of-life (EoL)** as explicit inputs in life cycle assessment (LCA) models of textile manufacturing processes.

The model was **developed and validated for knitting processes** — circular, seamless and hosiery — where machine masses, productivities and operating regimes have been cross-checked against OEM documentation. However, the framework is **process-agnostic by design**: the same two formulas (capital good allocation and EoL allocation) apply without modification to any textile manufacturing process where a machine mass and a throughput can be defined. Section 5 provides explicit guidance for adapting the model to spinning, weaving, cut & sew, dyeing and other processes.

Both the capital good allocation and the EoL allocation follow the same mass-based amortisation principle, consistent with EF guidelines (Fazio et al., 2020) and ecoinvent practice (Weidema et al., 2013).

## 2. Conceptual Basis

### 2.1 Why include machinery as a capital good?

Machinery is a long-lived asset whose environmental burden (production of the machine itself) should be distributed over all the functional units it delivers during its service life. Excluding it from the foreground model systematically underestimates the infrastructure contribution, particularly for heavy, energy-intensive machines with moderate throughput.

EF guidance requires including capital goods and their EoL **unless excluded under the 3% cut-off rule with explicit justification** [1]. ecoinvent implements the same logic by integrating infrastructure and its decommissioning into construction datasets, amortised over service delivered [3, 4].

### 2.2 What is modelled

For every 1 kg of knitted (or otherwise manufactured) output, the model attributes:

1. **Capital good (machine)**: kg of machine per kg of output — representing the share of the machine's mass "consumed" to produce 1 kg of product.
2. **End-of-life (machine)**: kg of each material fraction at EoL per kg of output — representing the decommissioning burden distributed over lifetime output, split by material type (steel vs. other).

Both quantities are expressed in **kg/kg** (mass of machine or material per kg of fabric), keeping the model neutral and compatible with any downstream impact method.

### 2.3 ecoinvent implementation

The capital good input is typically linked to the ecoinvent dataset:



**Dataset:** market for building machine / GLO

**Reference flow:** 1 unit (= 1 machine)

**Unit in foreground model:** unit/kg

**Conversion:** Amount (unit/kg) = machine mass (kg) / Q\_lifetime (kg) / machine mass (kg) = 1 / Q\_lifetime

Simpler: **Amount (unit/kg) = 1 / Q\_lifetime** — i.e. one machine amortised over its lifetime output.

## 3. Calculation Framework

### 3.1 Capital good allocation - core formula

The machine mass attributed per kilogram of fabric is:

$$M_{\text{machine/kg}} \left( \frac{\text{kg}_{\text{machine}}}{\text{kg}_{\text{fabric}}} \right) = \frac{M_{\text{machine}} (\text{kg})}{Q_{\text{lifetime}} (\text{kg})}$$

where:

$$Q_{\text{lifetime}} \text{ (kg)} = \text{productivity (kg/h)} \times \text{hours/day} \times \text{days/year} \times \text{years}$$

To express this in **ecoinvent unit/kg format**:

$$\text{Amount (unit/kg)} = \frac{M_{\text{machine/kg}}}{M_{\text{machine}}} = \frac{1}{Q_{\text{lifetime}}}$$

## 3.2 End-of-life allocation - core formula

The EoL flow for each material fraction  $i$  per kilogram of fabric is:

$$\text{EoL}_i \left( \frac{\text{kg}_i}{\text{kg}_{\text{fabric}}} \right) = \frac{M_{\text{machine}} \times \text{share}_i}{Q_{\text{lifetime}}}$$

where  $\text{share}_i$  is the mass fraction of material  $i$  in the machine (e.g. 0.85 for steel, 0.15 for other materials). The same  $Q_{\text{lifetime}}$  denominator is used for both the capital good and EoL allocations, ensuring internal consistency.

## 3.3 Parameter definitions

| Parameter    | Unit    | Definition and guidance                                                                                                                                                     |
|--------------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| M_machine    | kg      | Total machine mass from OEM brochure or datasheet. Includes frame, needles, drive system and electronics. Excludes consumables (yarn, lubricants).                          |
| productivity | kg/h    | Net output per operating hour. Taken from OEM specifications at nominal speed and standard conditions. Adjust for observed efficiency if factory data are available.        |
| hours/day    | h/day   | Operating hours per calendar day: 8 h (single shift), 16 h (double shift), 24 h (continuous).                                                                               |
| days/year    | days/yr | Production days per year, net of planned maintenance and seasonal shutdowns. Use 300 days/yr as default; 365 for continuous operations.                                     |
| years        | yr      | Reference service life. Range: 10 yr (WCS) to 20 yr (BCS), with 15 yr as average. Align with machine lifetime, not building lifetime.                                       |
| Q_lifetime   | kg      | Total lifetime output. Derived from the formula above. Common denominator for both capital good and EoL allocations.                                                        |
| share_i      | —       | Mass fraction of material $i$ in the machine. Default: 85% steel, 15% other materials (electronics, non-ferrous components). Update if OEM material breakdown is available. |

## 3.4 Scenario parameters



**Consistency rule:** keep days/year consistent within each scenario. Do not mix 300 days/yr (weekday operation) with 365 days/yr (continuous) in the same scenario. Report all four drivers (years, h/d, d/y, kg/h) alongside every result to make the allocation transparent and auditable.

The three standard scenarios used across all knitting process models:

| Scenario         | Years | h/day | days/yr | Q direction        | Allocation direction                    |
|------------------|-------|-------|---------|--------------------|-----------------------------------------|
| WCS (Worst Case) | 10    | 8     | 300     | Lowest Q_lifetime  | Highest kg/kg (most burden per kg)      |
| Average          | 15    | 16    | 300     | Mid Q_lifetime     | Mid allocation — <b>used as default</b> |
| BCS (Best Case)  | 20    | 24    | 365     | Highest Q_lifetime | Lowest kg/kg (least burden per kg)      |

## 4. Default Values by Knitting Technology

## 4.1 Circular knitting

**Machine reference:** Leadsfon SJ3.0 / KILM-84AV

**Machine mass:** 2,800 kg

**Productivity:** ~30 kg/h

**Source:** [Mayer & Cie.](#) — [Leadsfon](#)

| Scenario       | Years     | h/day     | days/yr    | Q_lifetime (kg)  | M_machine/kg (kg/kg) | ecoinvent (unit/kg) |
|----------------|-----------|-----------|------------|------------------|----------------------|---------------------|
| WCS            | 10        | 8         | 300        | 720,000          | 0.0039               | —                   |
| <b>Average</b> | <b>15</b> | <b>16</b> | <b>300</b> | <b>2,160,000</b> | <b>0.0013</b>        | <b>4.64E-07</b>     |
| BCS            | 20        | 24        | 365        | 5,256,000        | 0.0005               | —                   |

**EoL allocation (average scenario, 2,800 kg machine):**

| Material                                   | Share | Mass (kg) | EoL (kg/kg fabric) |
|--------------------------------------------|-------|-----------|--------------------|
| Steel                                      | 85%   | 2,380     | 0.0011             |
| Other materials (electronics, non-ferrous) | 15%   | 420       | 0.0002             |

**ecoinvent format — average scenario:**

**1 2 3 4** 2,800 kg machine → 0.0013 kg/kg fabric → 1 unit → **X = 4.64×10<sup>-7</sup> unit/kg**  
 Derivation:  $1 / Q_{\text{lifetime}} = 1 / 2,160,000 = 4.63 \times 10^{-7} \approx \mathbf{4.64E-07 \text{ unit/kg}}$

## 4.2 Flat knitting

**Machine reference:** Fengfan double-carriage computerised flat knitting machine (comparable to KARL MAYER STOLL CMS 503 ki L class)

**Machine mass:** 1,000 kg

**Productivity:** ~2.5 kg/h

**Source:** Fengfan / Fibre2Fashion supplier listings; Sino Finetex (2025); Changhua Knitting Machine benchmarks

| Scenario       | Years     | h/day     | days/yr    | Q_lifetime (kg) | M_machine/kg (kg/kg) | ecoinvent (unit/kg) |
|----------------|-----------|-----------|------------|-----------------|----------------------|---------------------|
| WCS            | 10        | 8         | 300        | 60,000          | 0.0167               | —                   |
| <b>Average</b> | <b>15</b> | <b>16</b> | <b>300</b> | <b>180,000</b>  | <b>0.0056</b>        | <b>2.8E-06</b>      |
| BCS            | 20        | 24        | 365        | 438,000         | 0.0023               | —                   |

**EoL allocation (average scenario, 1,000 kg machine):**

| Material                                   | Share | Mass (kg) | EoL (kg/kg fabric) |
|--------------------------------------------|-------|-----------|--------------------|
| Steel                                      | 85%   | 850       | 0.00472            |
| Other materials (electronics, non-ferrous) | 15%   | 150       | 0.00083            |

**ecoinvent format — average scenario:**

1,000 kg machine → 0.0056 kg/kg fabric → 1 unit (= 2,000 kg generic machine) → **X = 2.8×10<sup>-6</sup> unit/kg**

Derivation:  $0.0056 \div 2,000 = \mathbf{2.8E-06 \text{ unit/kg}}$

Note: the ecoinvent [market for building machine {GLO}](#) dataset uses a reference unit of ~2,000 kg. The flat knitting value therefore differs from the simpler  $1/Q_{\text{lifetime}}$  formula — use  $0.0056 \div 2,000$  directly.

## 4.3 Seamless knitting



**Machine reference:** Santoni SM8-TR1S

**Machine mass:** 750 kg

**Productivity:** ~26.2 kg/h

**Source:** [Santoni SM8-TR1S brochure](#) — [Textile World, ITMA 2007](#)

| Scenario       | Years     | h/day     | days/yr    | Q_lifetime (kg)  | M_machine/kg (kg/kg) | ecoinvent (unit/kg) |
|----------------|-----------|-----------|------------|------------------|----------------------|---------------------|
| WCS            | 10        | 8         | 300        | 628,800          | 0.0012               | —                   |
| <b>Average</b> | <b>15</b> | <b>16</b> | <b>300</b> | <b>1,886,400</b> | <b>0.0004</b>        | <b>5.30E-07</b>     |
| BCS            | 20        | 24        | 365        | 4,590,240        | 0.0002               | —                   |

**EoL allocation (average scenario, 750 kg machine):**

| Material                                   | Share | Mass (kg) | EoL (kg/kg fabric) |
|--------------------------------------------|-------|-----------|--------------------|
| Steel                                      | 85%   | 637.5     | 0.0003             |
| Other materials (electronics, non-ferrous) | 15%   | 112.5     | 0.0001             |

**ecoinvent format — average scenario:**



750 kg machine → 0.0004 kg/kg fabric → 1 unit → **X =  $5.30 \times 10^{-7}$  unit/kg**

Derivation:  $1 / Q_{lifetime} = 1 / 1,886,400 \approx 5.30E-07$  unit/kg

## 4.4 3D Knitting (WHOLEGARMENT)

**Machine reference:** Stoll CMS 3D / Shima Seiki SWG-XR (representative mid-range)

**Machine mass:** 1,200 kg

**Productivity:** ~5 kg/h

**Source:** Stoll / Karl Mayer catalogue; Shima Seiki SWG-XR datasheets (2025)

| Scenario       | Years     | h/day     | days/yr    | Q_lifetime (kg) | M_machine/kg (kg/kg) | ecoinvent (unit/kg) |
|----------------|-----------|-----------|------------|-----------------|----------------------|---------------------|
| WCS            | 10        | 8         | 300        | 120,000         | 0.0100               | —                   |
| <b>Average</b> | <b>15</b> | <b>16</b> | <b>300</b> | <b>360,000</b>  | <b>0.00333</b>       | <b>1.667E-06</b>    |
| BCS            | 20        | 24        | 365        | 876,000         | 0.00137              | —                   |

**EoL allocation (average scenario, 1,200 kg machine):**

| Material                                   | Share | Mass (kg) | EoL (kg/kg fabric) |
|--------------------------------------------|-------|-----------|--------------------|
| Steel                                      | 85%   | 1,020     | 0.00283            |
| Other materials (electronics, non-ferrous) | 15%   | 180       | 0.00050            |

**ecoinvent format — average scenario:**

1,200 kg machine → 0.00333 kg/kg fabric → 1 unit (= 2,000 kg generic machine) → **X =  $1.667 \times 10^{-6}$  unit/kg**

Derivation:  $0.00333 \div 2,000 = 1.667E-06$  unit/kg (also =  $1 / Q_{lifetime} = 1 / 360,000 = 2.78E-06$  — note: the two routes differ because the ecoinvent reference unit is 2,000 kg, not  $M_{machine}$ . Use  $M_{machine}/kg \div M_{machine}$  consistently.)

## 4.5 Hosiery knitting

**Machine reference:** Lonati GE616DF / GE516DF family  
**Machine mass:** 300 kg  
**Productivity:** ~10 kg/h (based on ~0.60 g per pair cycle)  
**Source:** [Lonati product family](#).

| Scenario       | Years     | h/day     | days/yr    | Q_lifetime (kg) | M_machine/kg (kg/kg) | ecoinvent (unit/kg) |
|----------------|-----------|-----------|------------|-----------------|----------------------|---------------------|
| WCS            | 10        | 8         | 300        | 240,000         | 0.0013               | —                   |
| <b>Average</b> | <b>15</b> | <b>16</b> | <b>300</b> | <b>720,000</b>  | <b>0.0004</b>        | <b>1.39E-06</b>     |
| BCS            | 20        | 24        | 365        | 1,752,000       | 0.0002               | —                   |

**EoL allocation (average scenario, 300 kg machine):**

| Material                                   | Share | Mass (kg) | EoL (kg/kg fabric) |
|--------------------------------------------|-------|-----------|--------------------|
| Steel                                      | 85%   | 255       | 0.00035            |
| Other materials (electronics, non-ferrous) | 15%   | 45        | 0.00006            |

**ecoinvent format — average scenario:**

**1 2 3 4** 300 kg machine → 0.0004 kg/kg fabric → 1 unit → **X = 1.39×10<sup>-6</sup> unit/kg**  
 Derivation:  $1 / Q_{\text{lifetime}} = 1 / 720,000 \approx 1.39\text{E-}06 \text{ unit/kg}$

## 4.6 Cross-technology summary

| Technology | Machine          | Mass (kg) | Productivity (kg/h) | Q_avg (kg) | M/kg avg (kg/kg) | ecoinvent avg (unit/kg) | EoL steel avg (kg/kg) |
|------------|------------------|-----------|---------------------|------------|------------------|-------------------------|-----------------------|
| Circular   | Leadsfon SJ3.0   | 2,800     | 30                  | 2,160,000  | 0.0013           | 4.64E-07                | 0.0011                |
| Seamless   | Santoni SM8-TR1S | 750       | 26.2                | 1,886,400  | 0.0004           | 5.30E-07                | 0.0003                |
| Hosiery    | Lonati GE616DF   | 300       | 10                  | 720,000    | 0.0004           | 1.39E-06                | 0.00035               |

**💡 Reading the table:** Circular knitting has the heaviest machine (2,800 kg) but also the highest throughput (30 kg/h), so the per-kg allocation is similar to hosiery (0.0013 vs 0.0004). Seamless also lands at 0.0004 kg/kg despite a lighter machine, because its throughput is proportionally lower. The ecoinvent unit/kg values reflect 1 machine amortised over Q\_lifetime — always verify by dividing 1 / Q\_lifetime directly.

## 5. Adapting the Model to Other Textile Processes

### 5.1 Why the model transfers directly

The two core formulas in Section 3 contain **no knitting-specific assumptions**. The only inputs are machine mass (M\_machine), productivity (kg/h), and operating regime (h/day, days/yr, years). These quantities exist and are measurable for every capital-intensive textile process. The logic is therefore identical whether the machine is a ring-spinning frame, a rapier loom, a jet dyeing vessel or a cutting table.

The reasons the model was first built and validated for knitting are practical rather than methodological: knitting machines have well-documented OEM brochures with explicit mass and productivity data, and a published benchmark (openLCA

Nexus organic cotton sweater) provided an independent cross-check. These same validation steps can and should be repeated for each new process.



**Key principle:** the formula does not change. What changes across processes are the *input values* — machine mass, productivity, and service life. The structure of the calculation, the scenario logic (WCS / Average / BCS), the ecoinvent dataset used, and the EoL material split all remain the same.

## 5.2 What to collect for a new process

| Step                    | What to collect                                                                                                                                                         | Typical source                                                                                                                                      |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Machine mass         | Total machine weight in kg, including frame, drive system and electronics. Exclude consumables (bobbins, needles, belts).                                               | OEM brochure, technical datasheet, or CE declaration of conformity (which often includes mass).                                                     |
| 2. Productivity         | Net output in kg/h at nominal operating conditions. For multi-head or multi-spindle machines, confirm whether the figure is per machine or per head.                    | OEM product page or brochure. Cross-check with published LCA literature (e.g. van der Velden et al. 2014 for spinning and weaving benchmarks).      |
| 3. Operating regime     | Hours/day and days/year for the target factory or the representative industry average. Keep consistent with the same scenario used for energy and water inventories.    | Factory data (primary) or industry benchmarks (secondary). Use 300 days/yr as default for batch/shift operations; 330–365 for continuous processes. |
| 4. Service life         | Expected machine lifetime in years. Align with the value used for building infrastructure if both are modelled.                                                         | OEM warranty documentation, industry surveys, or the ecoinvent default for the relevant machine category.                                           |
| 5. Material composition | Share of steel vs. other materials. If OEM breakdown is unavailable, the 85% steel / 15% other default is a reasonable starting point for most heavy textile machinery. | OEM environmental product declaration (EPD), material safety data, or World Steel Association LCI data [12].                                        |

## 5.3 Indicative reference ranges by process

The values below are **order-of-magnitude orientations** drawn from published LCA literature and equipment catalogues. They are not validated defaults — replace them with process-specific calculations using the steps above whenever possible.

| Process                      | Representative machine                          | Typical mass (kg)   | Typical productivity (kg/h) | Indicative M/kg avg (kg/kg) | Notes                                                                                                                                                                         |
|------------------------------|-------------------------------------------------|---------------------|-----------------------------|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ring spinning                | Ring frame (e.g. Rieter G 38)                   | 3,000 – 8,000       | 100 – 400                   | ~1E-03 – 5E-03              | High spindle count; mass and productivity both scale with frame length. Confirm per-frame or per-spindle basis.                                                               |
| Open-end (rotor) spinning    | Rotor frame (e.g. Rieter R 37)                  | 4,000 – 10,000      | 300 – 800                   | ~5E-04 – 2E-03              | Higher throughput than ring; lower M/kg for comparable frame mass.                                                                                                            |
| Rapier weaving               | Rapier loom (e.g. Picanol Rapido)               | 1,500 – 3,000       | 20 – 80                     | ~2E-03 – 1E-02              | Productivity highly dependent on fabric weight and width. Express as kg of woven fabric per hour.                                                                             |
| Airjet weaving               | Airjet loom (e.g. Toyota JAT810)                | 1,200 – 2,500       | 60 – 200                    | ~5E-04 – 3E-03              | Higher speed than rapier; lower M/kg. Airjet looms are lighter per unit output.                                                                                               |
| Cut & Sew (garment assembly) | Industrial sewing machine (e.g. Juki DDL-9000C) | 30 – 80 (head only) | 1 – 5                       | ~5E-03 – 5E-02              | Very light machines but very low throughput → relatively high M/kg. Include table/stand mass if significant. Consider modelling a full workstation (machine + table + motor). |
| Jet dyeing                   | Jet dyeing machine (e.g. Thies iMaster)         | 2,000 – 8,000       | 50 – 200 (batch)            | ~1E-03 – 1E-02              | Batch process: express productivity as kg of fabric per batch cycle, then                                                                                                     |

| Process | Representative machine | Typical mass (kg) | Typical productivity (kg/h) | Indicative M/kg avg (kg/kg) | Notes                                                                                       |
|---------|------------------------|-------------------|-----------------------------|-----------------------------|---------------------------------------------------------------------------------------------|
|         |                        |                   |                             |                             | convert to kg/h using average cycle time. Stainless steel dominant → adjust material split. |



These ranges span one order of magnitude within each process category. The spread reflects genuine variability in machine size, gauge, and operating regime — not uncertainty in the method. Always derive the actual value from specific machine data rather than reading from this table.

## 5.4 Process-specific considerations

**Spinning:** ring and rotor frames have many spindles or rotors; productivity figures in OEM literature are often given *per spindle* or *per rotor*, not per machine. Multiply by the number of spindles/rotors per frame before applying the formula. Service life is typically 20–30 years for spinning frames — longer than knitting machines.

**Weaving:** loom productivity is often reported in metres per minute (m/min) or picks per minute (PPM), not kg/h. Convert using: productivity (kg/h) = speed (m/min) × 60 × fabric width (m) × fabric areal weight (kg/m<sup>2</sup>). Reed width and fabric weight must be specified.

**Cut & Sew:** sewing machine mass is very low (30–80 kg for the head), but productivity is also very low (1–5 kg/h per workstation). The resulting M/kg is higher than for fabric-forming processes, but the absolute contribution to total LCA impact is typically small given the low machine mass. Consider whether to include the full workstation (machine head + table + motor + frame) for a more representative result.

**Dyeing and finishing:** many dyeing machines are made predominantly of stainless steel rather than carbon steel. Adjust the material split accordingly (e.g. 70% stainless steel, 20% carbon steel, 10% other). Use the ecoinvent dataset for stainless steel production in the EoL module.

**Multi-machine cells:** where multiple machines operate in a cell sharing common infrastructure (creel, conveyor, inspection table), allocate the shared equipment mass proportionally to output, or include it as a separate capital good entry.

## 6. Worked Numerical Example





### Example: Circular knitting — average scenario (step by step)

#### Inputs:

- Machine mass: 2,800 kg (Leadsfon SJ3.0)
- Productivity: 30 kg/h
- Hours/day: 16 h (double shift)
- Days/year: 300
- Service life: 15 years

#### Step 1 — Q\_lifetime:

$Q_{\text{lifetime}} = 30 \times 16 \times 300 \times 15 = 2,160,000 \text{ kg}$

#### Step 2 — Capital good allocation:

$M/\text{kg} = 2,800 / 2,160,000 = 0.001296 \text{ kg/kg} \approx 0.0013 \text{ kg/kg}$

#### Step 3 — ecoinvent unit/kg:

$\text{Amount} = 1 / 2,160,000 = 4.63 \times 10^{-7} \approx 4.64 \text{E-}07 \text{ unit/kg}$

( $= M/\text{kg} \div M_{\text{machine}} = 0.0013 / 2,800$  — same result)

#### Step 4 — EoL: steel (85%):

$\text{EoL}_{\text{steel}} = 2,800 \times 0.85 / 2,160,000 = 2,380 / 2,160,000 = 0.00110 \text{ kg/kg}$

#### Step 5 — EoL: other materials (15%):

$\text{EoL}_{\text{other}} = 2,800 \times 0.15 / 2,160,000 = 420 / 2,160,000 = 0.000194 \text{ kg/kg} \approx 0.0002 \text{ kg/kg}$

**Check:**  $\text{EoL}_{\text{steel}} + \text{EoL}_{\text{other}} = 0.0011 + 0.0002 = 0.0013 = M/\text{kg} \checkmark$

## 7. Implementation in LCA Software

### 6.1 Capital good input

| Field               | Entry                                                                                                     |
|---------------------|-----------------------------------------------------------------------------------------------------------|
| Input type          | Capital good                                                                                              |
| Dataset name        | market for building machine                                                                               |
| Geography           | GLO                                                                                                       |
| Reference flow unit | unit                                                                                                      |
| Amount              | Technology-specific value from Section 4 (e.g. 4.64E-07 unit/kg for circular, average)                    |
| Comment             | Machine capital good amortised over lifetime output. See Capital Goods & EoL Methodological Document v1.0 |

### 6.2 End-of-life inputs

EoL flows are added as separate inputs (one per material fraction):

| Material              | Suggested ecoinvent dataset                                 | Unit | Amount (circular, avg) |
|-----------------------|-------------------------------------------------------------|------|------------------------|
| Steel                 | market for scrap steel / GLO (or waste treatment for steel) | kg   | 0.0011                 |
| Other materials (E&E) | market for waste electric and electronic equipment / GLO    | kg   | 0.0002                 |

### 6.3 Standards alignment

- **EF 3.1 guidance [1, 2]:** capital goods and EoL included by default unless below 3% cut-off with explicit justification.
- **ecoinvent v3 [3, 4, 5]:** infrastructure amortised over service delivered; EoL integrated into construction datasets and distributed proportionally.
- **This model:** mirrors the ecoinvent logic in pure mass terms, independent of any impact characterisation method. The mass-based approach is neutral and auditable.

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## 8. References

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